

Food Hub Project

Post Graduate Program in Artificial Intelligence and Machine Learning

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Contents / Agenda

- Executive Summary
- Business Problem Overview and Solution Approach
- Data Overview
- EDA - Univariate Analysis
- EDA - Multivariate Analysis

Executive Summary

The analysis of FoodHub's order, operational, and customer rating data highlights clear trends in cuisine popularity, pricing behavior, operational efficiency, and revenue distribution. American and Japanese cuisines dominate the platform in both order volume and weekend demand, followed by Italian and Chinese cuisine types. Order frequency spikes significantly on weekends, with more than twice as many orders as weekdays.

Revenue is concentrated among a small set of high-performing restaurants. Four establishments—Shake Shack, The Meatball Shop, Blue Ribbon Sushi, and Blue Ribbon Fried Chicken—contribute roughly 30% of total gross revenue. Net revenue is estimated at \$6,166.

Operational performance appears consistently strong. Median preparation times (27 minutes) and delivery times (25 minutes) fall within predictable ranges, with distributions showing slight skewness but overall stability. Delivery times follow an approximately normal distribution centered between 20 and 30 minutes. Weekend operations achieve faster delivery speeds (median ~22 minutes) but with more variability, while weekdays show slower but more consistent delivery times (median ~28 minutes). Roughly 10% of orders exceed a 60-minute delivery time, highlighting an opportunity for targeted operational improvements.

Overall, FoodHub's data reveals strong weekend demand, concentrated revenue drivers, efficient operational performance, but significant gaps in rating completeness—over 700 unrated orders—which may distort customer satisfaction analysis. The insights point to opportunities to improve data completeness, evaluate outlier-prone cuisines, optimize slow or inconsistent delivery segments, and strengthen partnerships with top-performing restaurants.

Conclusions & Recommendations

Demand and Restaurant Strategy

- ❑ Prioritize and Support top-performing restaurants (Shake Shack, The Meatball Shop, Blue Ribbon Sushi, Blue Ribbon Fried Chicken) since they generate ~30% of revenue.
- ❑ Expand partnership in American and Japanese cuisines, which show highest order volumes and weekend demand.
- ❑ Evaluate Underperforming cuisines (Korean, Vietnamese, Mediterranean) for menu optimization or promotional support.

Pricing & Revenue Optimization

- ❑ Introduce targeted promotions for higher cost cuisines (Thai, French) to improve conversion while managing customer price sensitivity.
- ❑ Explore upsell or bundle options for moderately priced cuisines (~\$15 median) to lift average order value.
- ❑ Analyze outlier in cuisine pricing to ensure consistency and eliminate potential mispriced items.

Operational Efficiency

- ❑ Investigate the 10% of orders with delivery times > 60 minutes to identify bottlenecks.
- ❑ Focus on stabilizing weekend delivery variability by leveraging demand forecasting.
- ❑ Review preparation time variability in Thai cuisine (highest variability) and replicate consistency practices from Korean cuisine (lowest variability).

Conclusions & Recommendations

Customer Ratings and Experience

- ☐ Improve rating capture by setting up a customer loyalty program to reward survey participation.
- ☐ Monitor ratings of '3' as they indicate the most operational variability and possible customer DSAT.
- ☐ Use qualitative feedback since ratings show little correlation with time or cost, suggesting perception-based drivers.

Data & Analytical Improvements

- ☐ Investigate cuisine-specific outlines (cost and pre time) to identify data anomalies or true operational issues.
- ☐ Enhanced data completeness to strengthen reliability of future insights.
- ☐ Develop tracking and monitoring mechanism to monitor delivery consistency across weekdays and weekends.

Business Problem Overview and Solution Approach

FoodHub has collected detailed data from customer orders placed through its mobile app platform. The company aims to analyze this data to gain insights into restaurant popularity, customer preferences, and operational performance. The analysis focuses on key variables, including cuisine type, restaurant name, order frequency, preparation time, delivery time, and customer ratings.

Methodology:

1. Data Preprocessing
 - Review and clean the dataset by checking for missing values and duplicates.
 - Standardize data formats and ensure all variables have appropriate data types.
2. Exploratory Data Analysis (EDA)
 - **Univariate Analysis:** Examine individual variables to understand their distributions, patterns and characteristics.
 - **Multivariate Analysis:** Investigate relationships between two or more variables to identify correlations and patterns.
3. Statistical Analysis and Visualization
 - Apply statistical techniques to summarize key metrics.
 - Use visualizations such as histograms, boxplots and heatmaps to identify trends and relations.

The insights derived from the analysis will guide strategic decisions, including targeting specific customer segments, optimizing product offerings, improving operational efficiencies, and enhancing overall customer satisfaction.

Data Overview 1

- Q1: The dataset used for this analysis is stored in an Excel file (foodhub_order.xls) that includes 9 variables (columns) and 1,898 observations (rows).
- The data dictionary for this file is organized as follows:
 1. order_id: Unique seven-digit numeric ID
 2. customer_id: Unique numeric ID
 3. restaurant_name: Restaurant Name
 4. cuisine_type: Cuisine type ordered by the customer
 5. cost_of_the_order: Cost of the Order
 6. day_of_the_week: Day when the order was placed
 7. rating: Rating given by the customer
 8. food_preparation_time: Time taken by the restaurant to prepare the food (minutes)
 9. delivery_time: Time taken to deliver the food order (minutes)

Data Overview 2

- Q2: What are the datatypes of the different columns in the dataset?
 - Four variables have an integer data type.
 - Four variables have an object data type.
 - One variable have a float data type.

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1898 entries, 0 to 1897
Data columns (total 9 columns):
#   Column                Non-Null Count  Dtype
---  -
0   order_id              1898 non-null   int64
1   customer_id           1898 non-null   int64
2   restaurant_name       1898 non-null   object
3   cuisine_type          1898 non-null   object
4   cost_of_the_order     1898 non-null   float64
5   day_of_the_week       1898 non-null   object
6   rating                1898 non-null   object
7   food_preparation_time 1898 non-null   int64
8   delivery_time         1898 non-null   int64
dtypes: float64(1), int64(4), object(4)
memory usage: 133.6+ KB
```


Data Overview 3

- Q3: Are there any missing values in the data?
 - There are no missing values in this dataset.
- Q4: Check the statistical summary of the data. What is the minimum, average, and maximum time it takes for food to be prepared once an order is placed?

MIN: 20.00 minutes	***					
AVG: 27.37 minutes		order_id	customer_id	cost_of_the_order	food_preparation_time	delivery_time
MAX: 35.00 minutes	count	1898.00	1898.00	1898.00	1898.00	1898.00
	mean	1477495.50	171168.48	16.50	27.37	24.16
	std	548.05	113698.14	7.48	4.63	4.97
	min	1476547.00	1311.00	4.47	20.00	15.00
	25%	1477021.25	77787.75	12.08	23.00	20.00
	50%	1477495.50	128600.00	14.14	27.00	25.00
	75%	1477969.75	270525.00	22.30	31.00	28.00
	max	1478444.00	405334.00	35.41	35.00	33.00

- Q5: How many orders are not rated?
 - 736 orders don't have a customer rating (~38% of the total orders).

Univariate Analysis 1

A Univariate analysis was conducted for all nine variables in the dataset to understand their individual distributions, patterns, and characteristics. The techniques used in this analysis included:

1. **Summary Statistics** – to evaluate key measures such as mean, median, mode, range and standard deviation.
2. **Frequency Distributions** – to observe the occurrences of categorical values and identify common or dominant categories.
3. **Other Statistical Techniques** – such as visual inspections using histograms, boxplots and count plots to detect skewness, outliers, and overall data spread.

Variable	order_id	customer_id	restaurant_name	cuisine_type	cost_of_the_order	day_of_the_week	rating	food_preparation_time	delivery_time
Type	Categorical	Categorical	Categorical	Categorical	Numeric	Categorical	Numeric	Numeric	Numeric
Analysis	Uniqueness	Uniqueness	Freq counts	Freq counts	Histograms & Boxplots	Freq counts	Uniqueness & Freq counts	Histograms & Boxplots	Histograms & Boxplots
Summary Statistics	Yes	Yes	N/A	N/A	Yes	N/A	N/A	Yes	Yes

Univariate Analysis 2

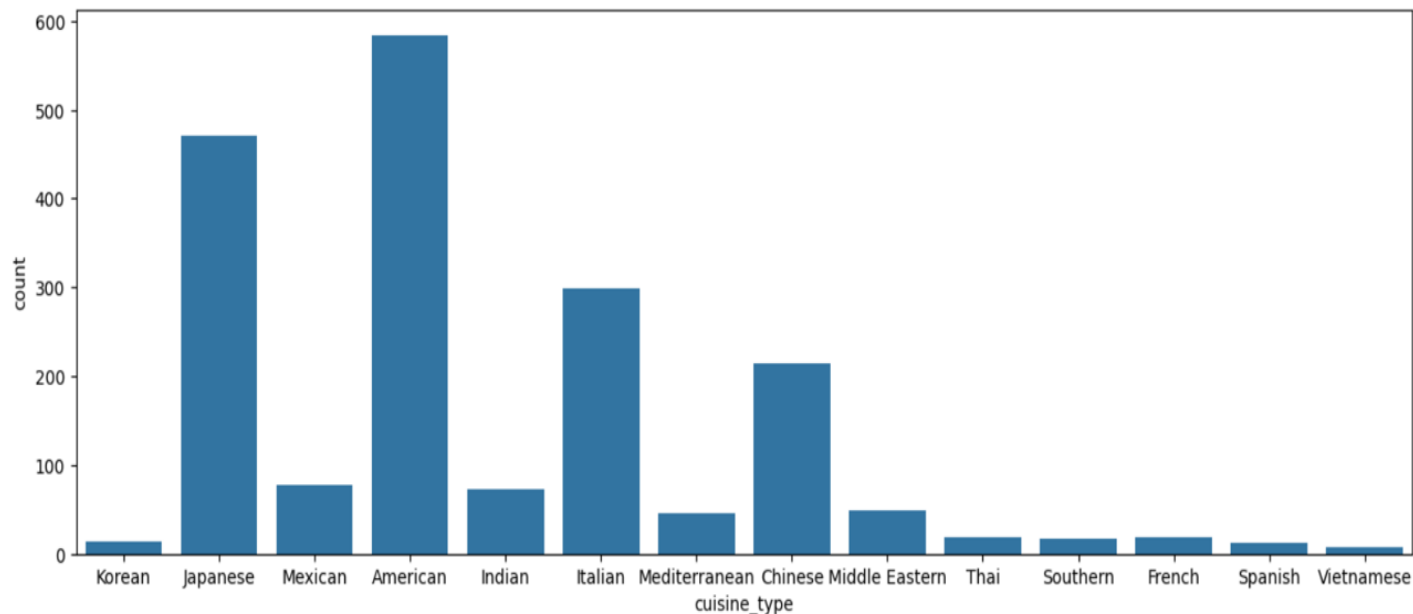
- Q6: Explore all the variables and provide observations on their distributions. (Generally, histograms, boxplots, countplots, etc. are used for univariate exploration.)
 - a) order_id: This variable contains 1898 unique values.
 - b) customer_id: This variable contains 1200 unique values.
 - c) restaurant_name: This variable contains 178 unique values. Some examples include: 'Hangawi', 'Blue Ribbon Sushi Izakaya', 'Cafe Habana', 'Blue Ribbon Fried Chicken', 'Dirty Bird to Go'.

```
... array(['Hangawi', 'Blue Ribbon Sushi Izakaya', 'Cafe Habana',  
         'Blue Ribbon Fried Chicken', 'Dirty Bird to Go',  
         'Tamarind TriBeCa', 'The Meatball Shop', 'Barbounia',  
         'Anjappar Chettinad', 'Bukhara Grill',  
         'Big Wong Restaurant \x8c\x94\x94', 'Empanada Mama (closed)', 'Pylos',  
         "Lucky's Famous Burgers", 'Shake Shack', 'Sushi of Gari',  
         'RedFarm Hudson', 'Blue Ribbon Sushi',  
         'Five Guys Burgers and Fries', 'Tortaria', 'Cafe Mogador',
```

Univariate Analysis 3

d) cuisine_type: This variable contains 14 unique values

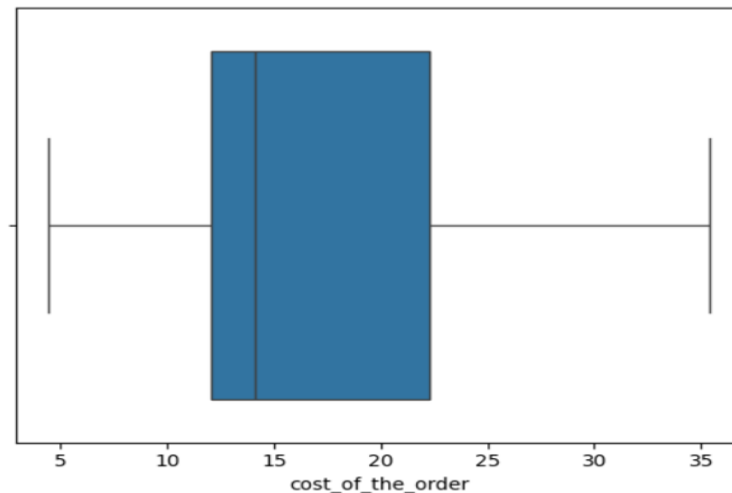
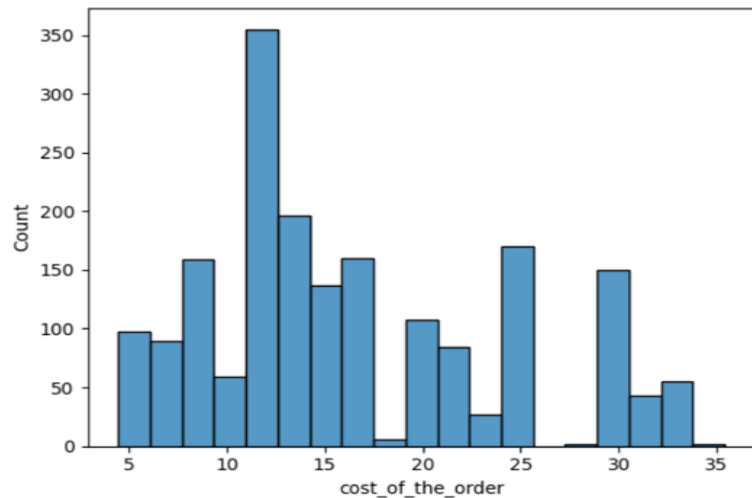
- Among all cuisine types, American and Japanese restaurants show the largest representation, with Italian and Chinese restaurants ranking next in frequency.



Univariate Analysis 4

e) cost_of_the_order:

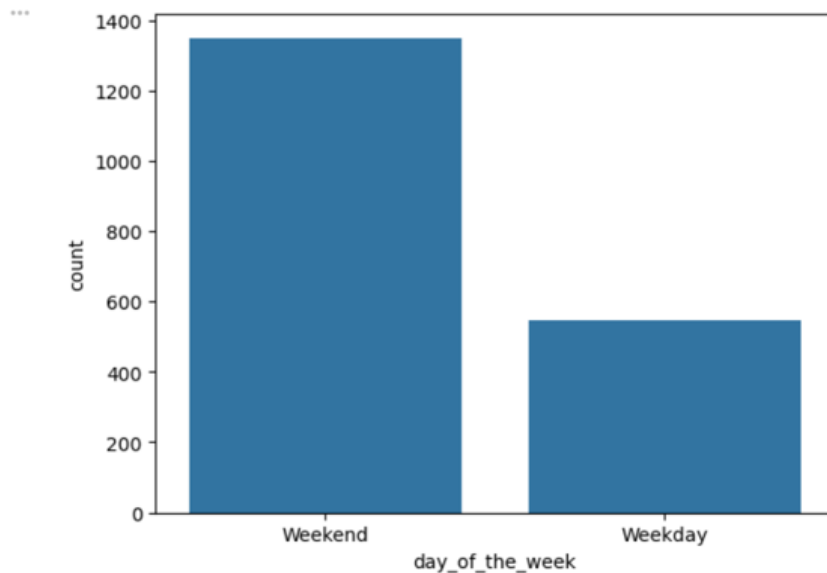
- 50% of the cost order values are less than \$15.
- Cost of the order does not have any outliers.
- The distribution of cost of order is highly right skewed - this indicates that most orders are moderately priced, but some can be quite expensive.



Univariate Analysis 5

f) day_of_the_week:

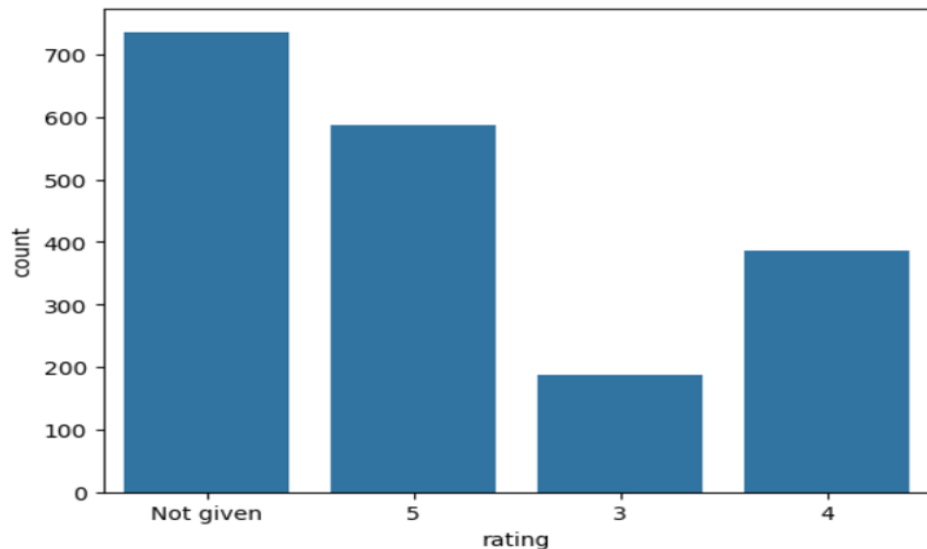
- The number of orders is more than double on weekends than on weekdays.



Univariate Analysis 6

g) rating:

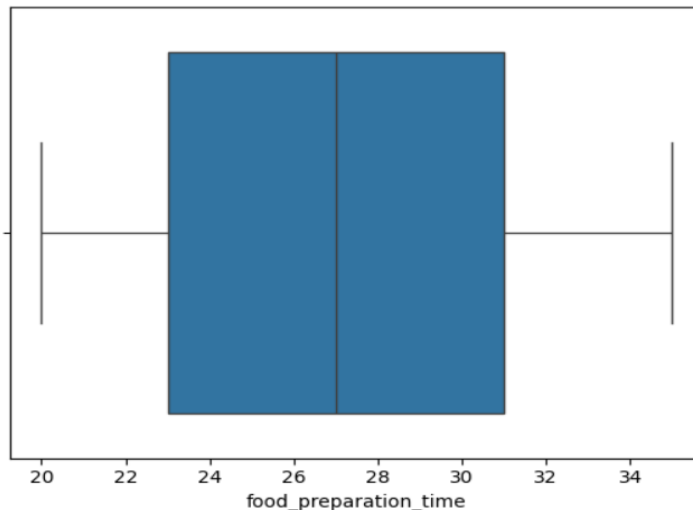
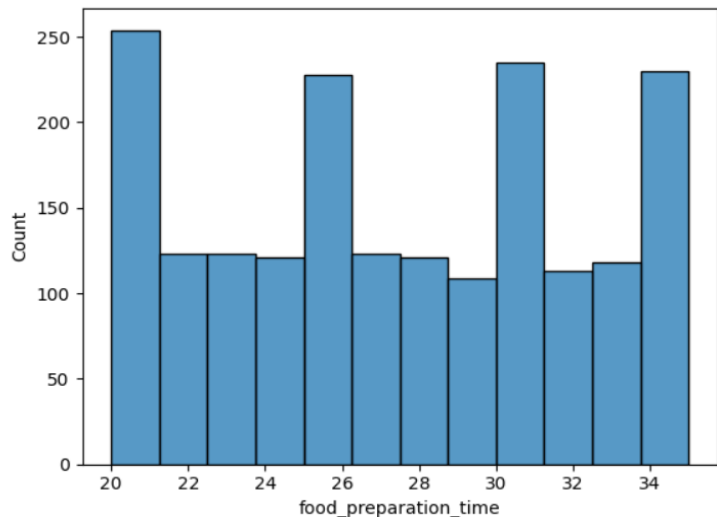
- Significant portion of rating counts (700) lack a rating.
- Satisfaction levels appear high (5) but the absence of rating counts may skew the analysis.



Univariate Analysis 7

h) food_preparation_time

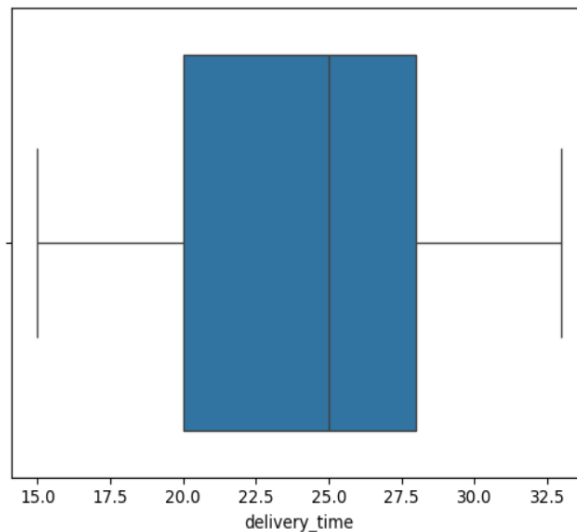
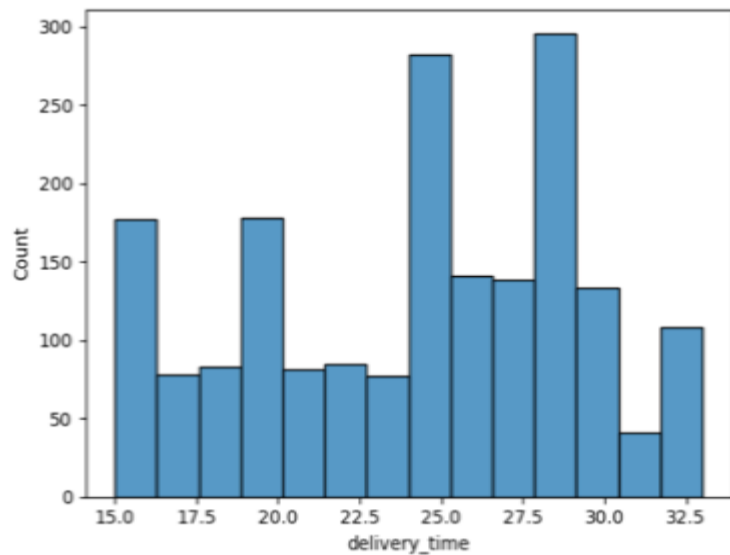
- 50% of the orders are prepared in less than 27 minutes.
- The distribution of the food preparation time histogram is bimodal with two peaks at 20 minutes and 30 minutes.
- The distribution of preparation times shows a slight positive skew, indicating that approximately 25% of orders (the 75th percentile) have preparation times ranging from 27 to 31 minutes.



Univariate Analysis 8

i) delivery_time

- 50% of the orders are delivered in less than 25 minutes.
- The delivery time histogram shows an approximately normal distribution, with most orders concentrated around 20 to 30 minutes.



Univariate Analysis 9

- Q7: Which are the top 5 restaurants in terms of the number of orders received?
 - The top five restaurants with the highest number of orders are Shake Shack, The Meatball Shop, Blue Ribbon Sushi, Blue Ribbon Fried Chicken, and Parm.

restaurant_name	count
Shake Shack	219
The Meatball Shop	132
Blue Ribbon Sushi	119
Blue Ribbon Fried Chicken	96
Parm	68

Univariate Analysis 10

- Q8: Which is the most popular cuisine on weekends?
 - The most popular weekend cuisine is American with 415 orders, followed by Japanese with 335.

cuisine_type	count
American	415
Japanese	335
Italian	207
Chinese	163
Mexican	53

Univariate Analysis 11

- Q9: What percentage of the orders cost more than 20 dollars?
 - The number of total orders that cost above 20 dollars is: 555.
 - Percentage of orders above 20 dollars: 29.24%.
- Q10: What is the mean order delivery time?
 - The mean delivery time for this dataset is 24.16 minutes.
- Q11: The company has decided to give 20% discount vouchers to the top 5 most frequent customers. Find the IDs of these customers and the number of orders they placed

count	
customer_id	
52832	13
47440	10
83287	9
250494	8
259341	7

Multivariate Analysis 1

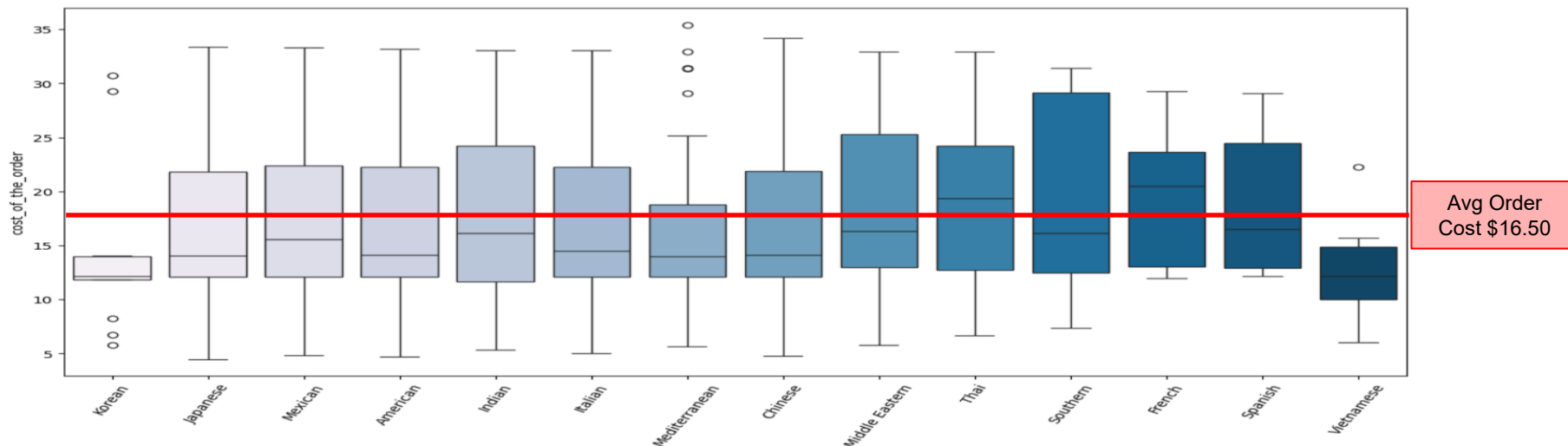
A Multivariate analysis was conducted to examine the relationships between two or more variables in the dataset to identify their patterns and correlations. The techniques used in this analysis included:

1. **Correlation Analysis** – to measure the strength of relationships.
2. **Heatmaps** – to visualize the relationships and patterns across multiple variables.
3. **Boxplots** – to compare measures of central tendency across multiple variables.

Relation	Cost of the Order and Cuisine Type	Food Preparation Time and Cuisine Type	Day of the Week and Delivery Time	Rating and Delivery Time	Rating and Food Preparation Time	Rating and Cost of the Order	Cost of the Order, Food Preparation Time, and Delivery Time
X	Cost of the Order (Numeric)	Cuisine Type (Categorical)	Day of the Week (Categorical)	Rating (Numeric)	Rating (Numeric)	Rating (Numeric)	All (Numeric)
Y	Cuisine Type (Categorical)	Food Preparation Type (Numeric)	Delivery Time (Numeric)	Delivery Time (Numeric)	Food Preparation Time (Numeric)	Cost of the Order (Numeric)	All (Numeric)
Analysis	Boxplot	Boxplot	Boxplot	Pointplot	Pointplot	Poinplot	Heatmap & Correlation

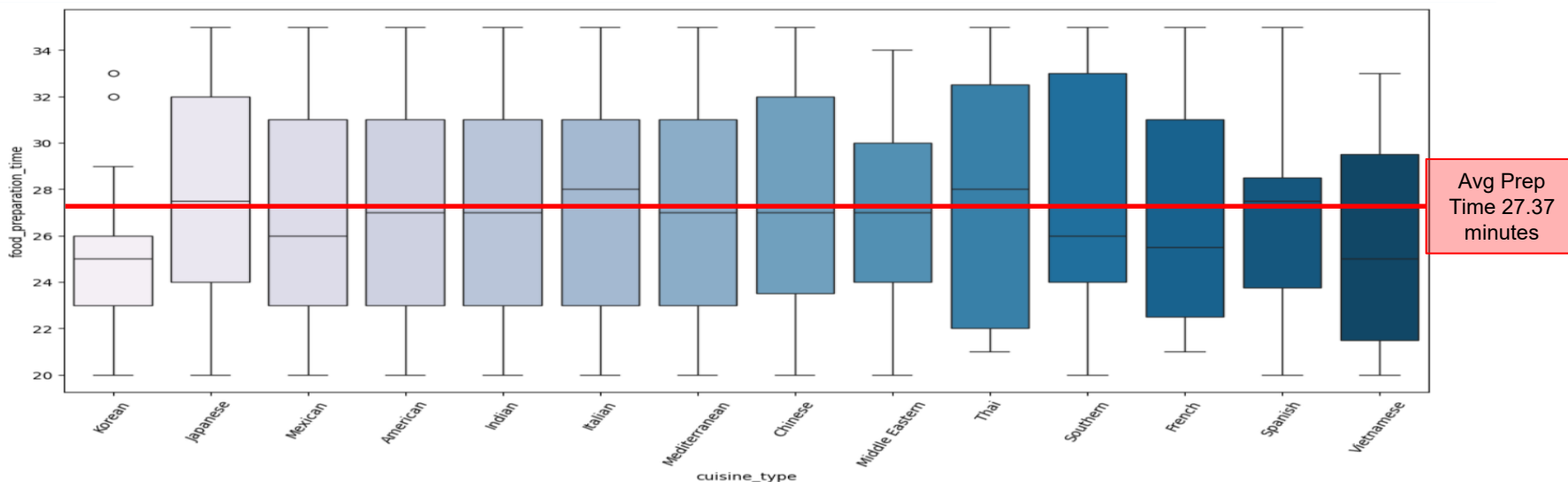
Multivariate Analysis 2

- Q12: Relation between Cost of the Order and Cuisine Type
 - Thai and French cuisine have the highest median cost at ~\$19 -20 which could indicate a left skewed distribution considering the average cost is \$16.50 (Median > Avg).
 - American and Japanese cuisine (most order counts) have very similar boxplots with a median cost at ~\$15.
 - Korean and Vietnamese cuisine (lowest order counts) have the lowest median cost at ~\$12.
 - Korean, Mediterranean and Vietnamese have the presence of outliers that should be further analyzed.



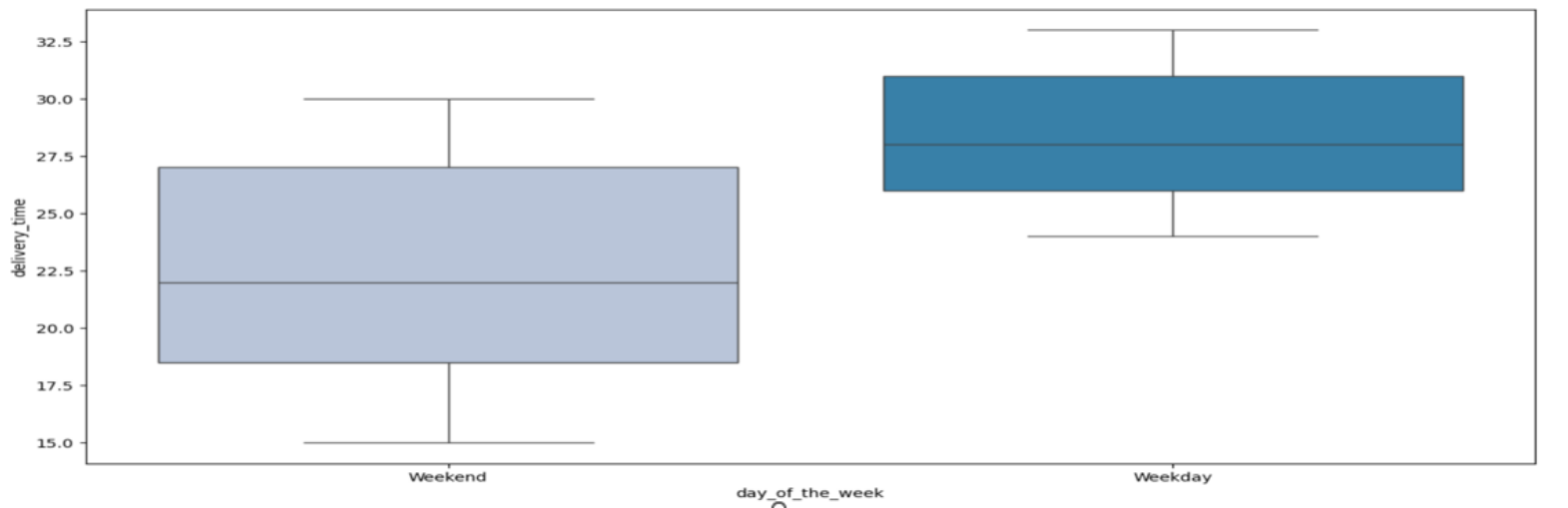
Multivariate Analysis 3

- Q12: Relation between Food Preparation Time and Cuisine Type
 - Korean, French and Vietnamese cuisine have the lowest median preparation time at ~25 minutes which could indicate a right skewed distribution considering the average preparation time is 27.37 minutes (Median<Avg).
 - American and Japanese cuisine (most order counts) have very similar boxplots with a median preparation time at ~27 minutes – Both cuisine types show moderate prep time variability.
 - Korean cuisine shows the least prep time variability while Thai cuisine shows the most prep time variability.



Multivariate Analysis 4

- Q12: Relation between Day of the Week and Delivery Time
 - Weekend median delivery time has the lowest delivery time at ~22 minutes with a wide-spread which indicates high variability and less consistency.
 - Weekday median delivery times has the highest delivery time at ~28 minutes with a narrow-spread which indicates low variability and more consistency.



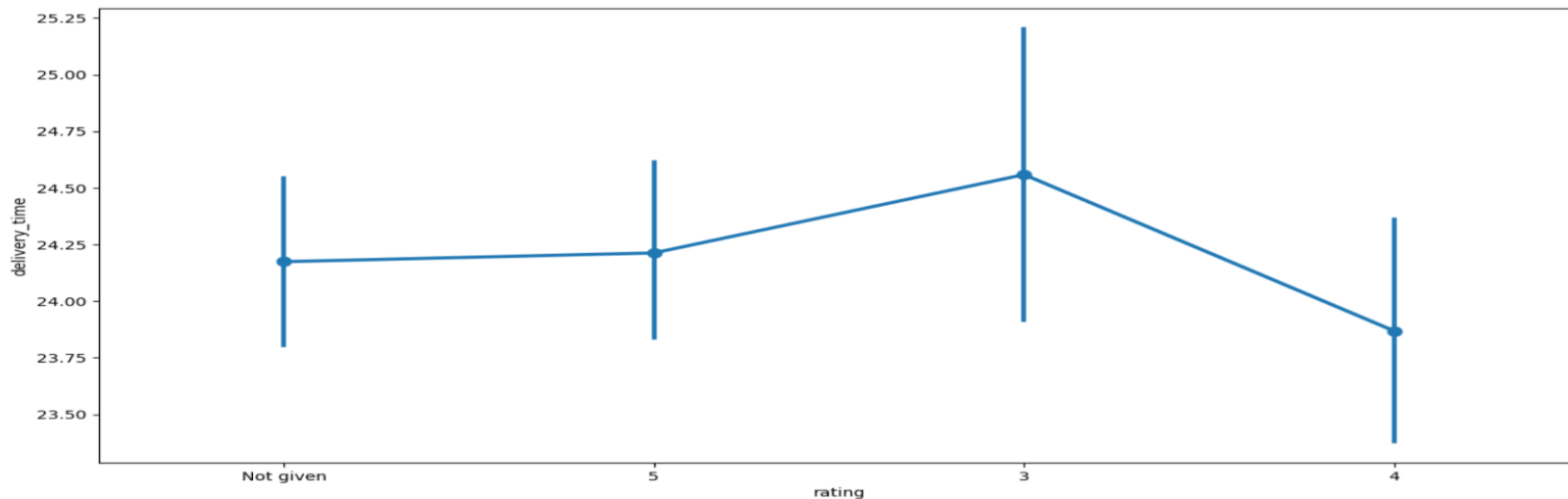
Multivariate Analysis 5

- Q12: Revenue by Restaurants
 - Total gross revenue: \$31,314.
 - Four restaurants (Shake Shack; The Meatball Shop; Blue Ribbon Sushi and Blue Ribbon Fried Chicken) make ~30% of the total gross revenue.
 - Two American cuisine restaurant (Shake Shack & Blue Ribbon Fried Chicken) make 16% of the total gross revenue.

restaurant_name	cost_of_the_order
Shake Shack	3579.53
The Meatball Shop	2145.21
Blue Ribbon Sushi	1903.95
Blue Ribbon Fried Chicken	1662.29
Parm	1112.76
RedFarm Broadway	965.13
RedFarm Hudson	921.21
TAO	834.50
Han Dynasty	755.29
Blue Ribbon Sushi Bar & Grill	666.62
Rubirosa	660.45
Sushi of Gari 46	640.87
Nobu Next Door	623.67
Five Guys Burgers and Fries	506.47

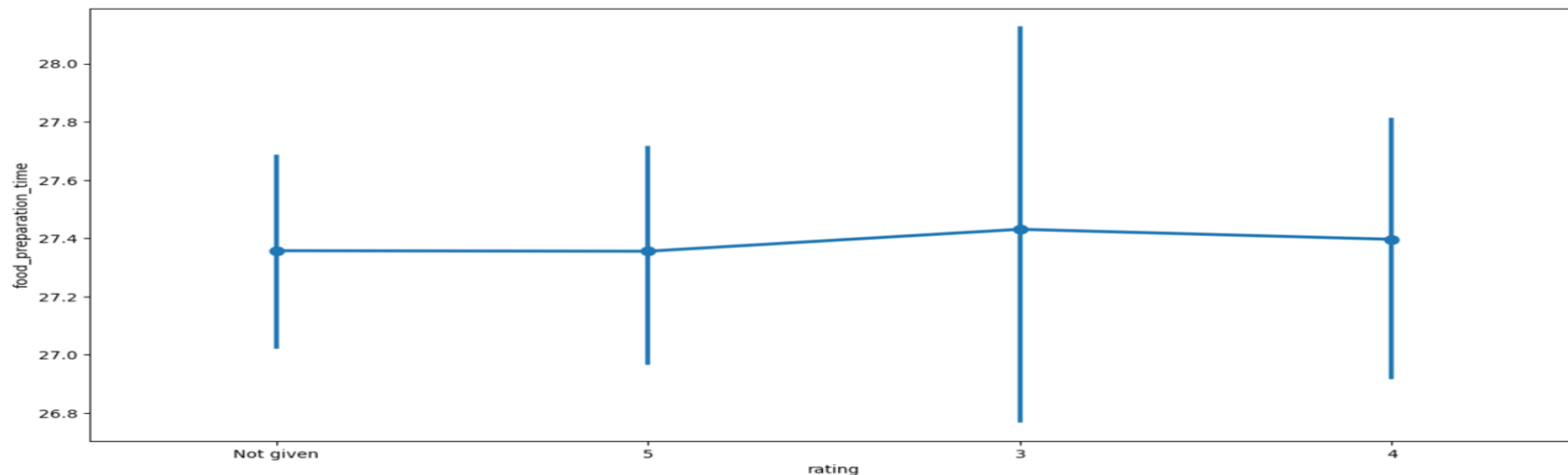
Multivariate Analysis 6

- Q12: Relation between Rating and Delivery Time
 - Rating '3' has the longest average delivery time and the widest spread suggesting high variability and less consistency.
 - Rating '5' and 'Not Given' show similar spreads with an almost identical average delivery time.
 - Rating '4' shows the lowest average delivery time which indicates customers propensity to reward short delivery times.



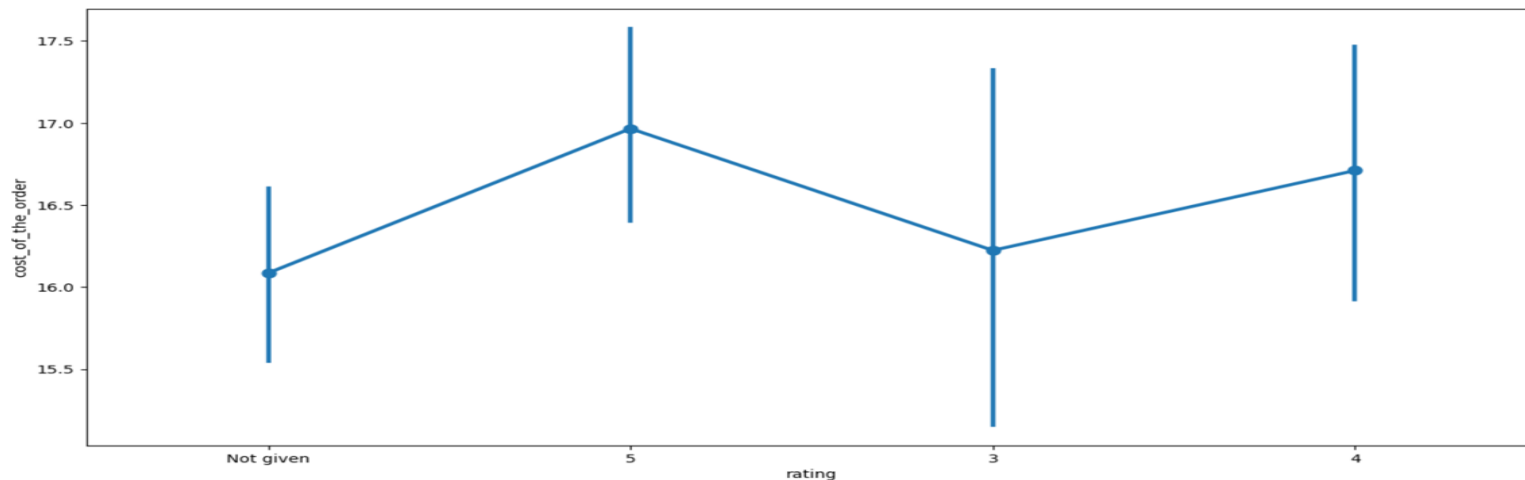
Multivariate Analysis 7

- Q12: Relation between Rating and Food Preparation Time
 - Rating '3' has the longest food preparation time and the widest spread suggesting high variability and less consistency.
 - Rating '5' and 'Not Given' show similar spreads with an almost identical food preparation time.
 - The average food preparation time is consistent across all rating categories which indicates no correlation between rating and food preparation time



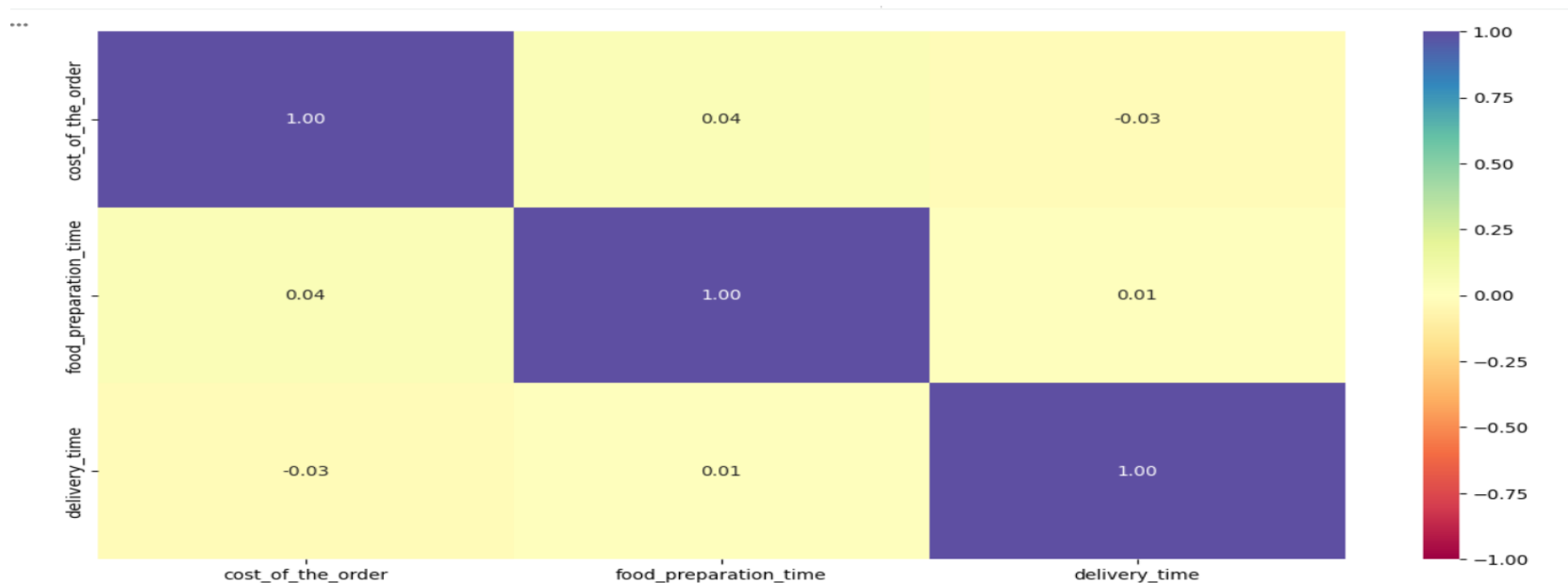
Multivariate Analysis 8

- Q12: Relation between Rating and Cost of the Order
 - Rating '3' has the longest order cost range and the widest spread suggesting high variability and less consistency.
 - Rating '5' is associated with the highest order cost which indicates no correlation between cost and rating – value perception matters more than price alone.



Multivariate Analysis 9

- Q12: Correlation among Cost of the Order, Food Preparation Time, and Delivery Time
 - There is no meaningful correlation across any combinations of these 3 variables
 - Metrics are largely uncorrelated which indicates that optimizing on variable will not cause any effect on the other variable



Multivariate Analysis 10

- Q13: Promotional Offer: restaurants with a rating count of 50 and an average rating greater than 4
 - Four restaurants (The Meatball Shop, Shake Shack, Blue Ribbon Sushi and Blue Ribbon Fried Chicken) have a rating of 4 and a rating count greater than 50.

	restaurant_name	rating
0	Shake Shack	133
1	The Meatball Shop	84
2	Blue Ribbon Sushi	73
3	Blue Ribbon Fried Chicken	64
4	RedFarm Broadway	41

	restaurant_name	rating
0	The Meatball Shop	4.511905
1	Shake Shack	4.278195
2	Blue Ribbon Sushi	4.219178
3	Blue Ribbon Fried Chicken	4.328125

Multivariate Analysis 11

- Q14: Company Net Revenue: Net revenue generated by the company across all orders
 - The Net Revenue is around 6166.3 dollars.

	order_id	customer_id	restaurant_name	cuisine_type	cost_of_the_order	day_of_the_week	rating	food_preparation_time	delivery_time	Revenue
0	1477147	337525	Hangawi	Korean	30.75	Weekend	Not given	25	20	7.6875
1	1477685	358141	Blue Ribbon Sushi Izakaya	Japanese	12.08	Weekend	Not given	25	23	1.8120
2	1477070	66393	Cafe Habana	Mexican	12.23	Weekday	5	23	28	1.8345
3	1477334	106968	Blue Ribbon Fried Chicken	American	29.20	Weekend	3	25	15	7.3000
4	1478249	76942	Dirty Bird to Go	American	11.59	Weekday	4	25	24	1.7385

Multivariate Analysis 12

- Q15: Total Time Required to Deliver: % of orders that take more than 60 minutes to get delivered from the time the order is placed (preparation time + delivery time)
 - The total number of orders with a delivery time greater than 60 minutes: 200.
 - Percentage of orders with a delivery time above 60 minutes: 10.54%.
- Q16: Mean Delivery Times: how does the mean delivery time vary during weekdays and weekends?
 - The mean delivery time on weekdays is around 28 minutes.
 - The mean delivery time on weekends is around 22 minutes.
- Q17: What are your conclusions from the analysis? What recommendations would you like to share to help improve the business?
 - Refer to slides 3, 4 & 5.



Happy Learning !

